ASHRAY SAXENA

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EDUCATION

Birla Institute of Technology and Science (BITS), Pilani · Pilani Campus 2023 – 2027
B.E. (Hons.) in Mechanical Engineering · currently in 4th Year, 1st Semester **CGPA: 8.48/10**

Amity International School, Noida 2023
Higher Secondary School Certificate, Central Board of Secondary Education (CBSE) **97.2%**

Relevant coursework: Gas Dynamics; Prime Movers & Fluid Machines; Fluid Mechanics; Heat Transfer; Applied Thermodynamics; Engineering Optimization; Advanced Mechanics of Solids; Vibrations & Control; Mechanisms & Machines; Computer-Aided Design; Probability & Statistics; Mathematics I–III.

Technical skills:

Machine learning	PyTorch, scikit-learn, Llama, PointNet++, YOLOv8, Gaussian processes, conformal prediction, active learning
Simulation & aero	SU2, ANSYS Fluent, OpenFOAM, COMSOL, Star-CCM+, XFOIL, XFLR5, OpenVSP, AeroSandbox, UNICADO
Methods	CasADi/IPOPT collocation, NSGA-II, POD/DMD, Dempster–Shafer theory, MSCKF/VIO, square-root filtering
Experimentation	Wind Tunnel Testing, Photo-Optics, CNC Machining, Pilot (UAVs), Manufacturing, Hydromechanics, Robotics
Code & embedded	Python, C, C++, MATLAB, Git, ROS2, STM32, TMS570, KiCad, Altium
CAD	SolidWorks, Siemens NX, CATIA V5, Fusion 360, PTC Creo, Onshape, AutoCAD

ACADEMIC RESEARCH EXPERIENCE

RWTH Aachen University · Institute of Aerospace Systems (ILR) Project ATLAS Aug 2026 – Feb 2027
Bachelor's Thesis: Credibility Estimation of Future Aircraft Concepts · Project ATLAS (DLR & Airbus) · Advisor: Univ.-Prof. Dr.-Ing. Eike Stumpf
Developing a credibility-assessment methodology for 2040–45 entry-into-service aircraft concepts and implementing it in the institute's preliminary-design environment UNICADO. Dempster–Shafer evidence theory over epistemic uncertainty in component mass estimation, with an expert-judgement reliability index for partner-supplied data; applied to kerosene and hydrogen ATLAS configurations.

University of Manchester · Mechanical, Aerospace & Civil Engineering Jul 2026 – Dec 2026
Visiting Research Intern: Novel Atomiser for High-Speed Propulsion Systems · Advisor: Dr. Kuppuraj Rajamanickam
Design of an atomiser concept for high-speed propulsion, coupling spray and primary-breakup physics to injector geometry to improve mixing uniformity and combustion stability.

RANS-Verified Surrogate Design of Wing Sections for a uSTOL Aircraft DOI 2026
Under review, *IAAI-27 (Innovative Applications of Artificial Intelligence) · first author · Advisor: Dr. Sharad Shrivastava, BITS Pilani*
End-to-end inverse-design pipeline showing that training-data generation, not surrogate architecture, dominates design quality: physics-regularised CST fit of the UIUC corpus, an NSGA-II Reynolds-envelope data engine, and a Reynolds-conditioned PointNet++ surrogate with a pressure-to-coefficient consistency loss. Collapses the train–test generalisation gap **9.8×** → **1.4×** and improves L/D by **15.4%** over a Bézier-GAN baseline, verified against SU2 RANS.

3D-Printed Rotary Cam Switch for Steering-Mode Transitions in a Four-Wheel Independent Steering Rover 2026
Submitted, *ACDSA 2027 (Int. Conf. on AI, Computer, Data Sciences and Applications) · V. Shah, A. Saxena, M. S. Dasgupta*
Low-cost state-transition mechanism for a four-wheel independent steering (4WIS) steer-by-wire rover: a 3D-printed rotary cam switch and encoder switch reliably between zero-radius pivot and Ackermann modes; tested on an autonomous rover.

Continuous-Wave Rotor Pressure Exchanger for Supercritical CO_2 Cycles · ISHRAE Jan 2026 – present
Advisor: Dr. Mani Sankar Dasgupta, BITS Pilani
Experimental study of transient pressure-wave management in rotor channels for direct-contact pressure exchange, achieving a **49% increase in isentropic efficiency** over conventional rotary-blade designs with reduced mixing losses.

INDEPENDENT RESEARCH EXPERIENCE

Coupled Dispersion of a Two-Flexible-Plate Structural–Acoustic Waveguide with Low-Mach Poiseuille Flow DOI Jan 2026 – present
Manuscript *in preparation for the Journal of Fluid Mechanics · sole author*
Closed-form $O(M)$ correction to every coupled branch by asymptotics in the fluid-loading parameter (Pridmore–Brown equation with plate-admittance boundaries); critical-layer threshold $M_{\text{crit}} = \Omega/\xi$; discrete wall modes shown not to be absorbed by the critical layer. Three independent numerical methods agree to **~1 part in 10^9**.

Joint Optimisation of Active Aerodynamics and Electric Energy Deployment for Minimum Lap Time in 2026 Formula One DOI 2026
Under review, *Proc. IMechE Part D: Journal of Automobile Engineering · sole author*
Minimum-time optimal control of a 2026 qualifying lap co-scheduling active-aerodynamic mode with **350 kW** MGU-K deploy/harvest under the regulatory rampdown (CasADi/IPOPT direct collocation; free racing line over true track limits). Three vehicle models from point-mass to transient single-track; calibrated to demonstrated capability, the model matches real 2026 poles to **within 0.17%**.

Frequency-Domain Analysis of Thermoacoustic Coupling in Liquid-Propellant Rocket Engines 2026
Accepted, *IEEE Aerospace Conference 2026*
Axisymmetric Helmholtz model (COMSOL) swept over the ignition band; Rayleigh-criterion mitigation by spatially decoupling pressure antinodes from the heat-release zone, giving a fast geometry and damping screen ahead of coupled reacting-flow simulation.

Rapid Aerodynamic Optimization of Aerospace Structures: Engine Nacelles 2026
Accepted, *Acceleron Aerospace Journal, Vol. 6 · first author*
CST (Kulfan) parameterisation of nacelle lip and cowl with an in-house viscous–potential solver and an automated lip-morphing optimizer; localises drag minima and incipient separation, resolving the non-monotonic drag-versus-flare trade at a fraction of RANS cost. A convergence self-audit established that the residual C_d sawtooth is numerical rather than physical.

PROFESSIONAL EXPERIENCE

LAT Aerospace · Aeromodelling & Avionics Intern; Gas Turbine Engine Team Dec 2025 – present

- **IRIS GNSS-denied navigation simulator.** Built the simulation and characterisation instrument for IRIS, a GNSS-denied visual–inertial navigation system: analytic trajectory generation, an Allan-variance-matched bit-faithful synthetic IMU, photoreal camera rendering (Unreal Engine + Cesium), and a browser-based flight console that runs the production MSCKF estimator (OpenVINS fork) blind and grades tracking, drift and divergence with NEES consistency tests. **0.12–0.2 m drift** on the 60 m canary trajectory; **sub-metre error** on photoreal imagery. Working from the filter internals (MSCKF nullspace projection and sliding-window InEKF on a composite $SE_k(3)$ state), derived a **square-root (√VINS / SR-ISWF)** backend formulation and identified its exact integration point in OpenVINS; designed a low-cost visual–inertial rig (webcam, ICM-42688-P IMU, ESP32 hardware timestamping) to benchmark it against the OpenVINS baseline.
- **SkyNET high-lift airfoil framework.** Inverse-design framework synthesising high-lift airfoils for blown-wing UAS from target lift and drag, evaluated with viscous–inviscid panel methods and SU2; designed single-slotted Fowler-flap and Coandă devices with deployment kinematics, using the slot-height-to-curvature ratio h/R to keep the blown jet attached.
- **FADEC & engine simulation.** DOI Modular dual-channel Full-Authority Digital Engine Controller for microturbojets (STM32H743 and TMS570LS1224, 8-layer board, torch-igniter control) with a C++ software-in-the-loop turbojet simulator; contributed to a hybrid powertrain pairing microturbojets with electric ducted fans on a **130 kg aircraft**.

DHIO Research & Engineering · CFD Engineering Intern May 2025 – Jul 2025
High-fidelity mesh-free CFD of liquid-propellant rocket-engine flows; frequency-domain analysis of thermoacoustic instability and chamber acoustic coupling (the seed of the IEEE AeroConf paper above); transient moving-mesh (ALE) study of an actuated drag-reduction system quantifying closed-versus-open aerodynamic loads.

India Space Lab · Satellite Communication Intern Jul 2025 – Aug 2025
Communication subsystem for CubeSat-class nanosatellites (link-budget optimisation, signal integrity); developed and deployed a telemetry, tracking and command (TT&C) system and supported transponder and antenna-array integration.

CrazyGoldFish.ai · Robotics Development Intern Oct 2024 – Feb 2025
Developed a page scanner and page-turning mechanism with a backend OCR and text-evaluation pipeline.

SELECTED PROJECTS

M-SWA: Multi-Modal Semantic Workbench Assistant · VLM-driven robotic arm Sep 2026 – present
Voice-driven workbench assistant in development: a vision-language model grounds spoken object references, which the arm points at with a laser, grasps and hands over, keeping a memory of objects seen; thermal and acoustic sensing answer physical questions (hot liquids, an overheating or noisy motor). Seeed SO-ARM101 Pro leader–follower arm with XIAO ESP32-S3 boards.

Uncertainty-Calibrated Transonic Flutter-Boundary Prediction · actively-sampled reduced-order model Jan 2026 – present
p–k/Theodorsen baseline plus coupled SU2 fluid–structure interaction feeding a Gaussian-process operator with split-conformal calibration and active learning; reproduces the AGARD 445.6 flutter speed index **within ~12%** and the Isogai Case-A transonic dip, converging the boundary in materially fewer solves than random sampling. Open pipeline, **88 passing tests**.

Aerodynamic Characterisation of STOL Configurations · IIT Bombay Nov 2025 – Jan 2026
Evaluated counter-circulation and distributed-integrated-propulsion blown-wing architectures; C_L–α polars quantifying stall delay, boundary-layer-control lift enhancement and propeller-slipstream/wing interaction in low-speed flight.

Czerny–Turner Raman Spectrophotometer · DST NIDHI PRAYAS May 2024 – Nov 2024
Advisor: Dr. Surojit Pande, BITS Pilani
High-precision Raman spectrophotometer (Czerny–Turner monochromator, laser optics) for non-destructive detection of organic biosignatures; optimised optical throughput and signal-to-noise across diffraction gratings and CCD selection.

LEADERSHIP

Team Captain · CRISS Robotics, Space Engineering & Robotics Team, BITS Pilani Jan 2024 – present

- Direct **50+ engineers** across four subsystems on a **$20,000+** budget; led the first Indian delegation to qualify for the Australian Rover Challenge 2026.
- Designed the rover chassis (CNC-machined and welded Al 6061), wheel assembly, actuators, pivot mechanism and robotic arm, with CNC-machined keys locking the C-arms to the wheel assembly; pivot steering with dynamic Ackermann geometry and a 3D-printed rotary cam switch for 4WIS mode transitions.

- Cut rover mass from **78 kg to 51 kg** by printing about half the connecting parts in PETG/Nylon and laser-cutting the covers; validated designs in ANSYS, COMSOL and SolidWorks, with CAD in 3DEXPERIENCE, Fusion 360 and CATIA V5.
- Integrated ROS with OpenCV for an autonomous camera-calibrated spectroscope; designed widefield-imaging and IR-spectrometer payloads and the rover sample-collection system.

CTO & Co-Founder · AtmosClear Pvt Ltd — Dec 2024 – Feb 2026

Developed and built SAMAP, a Smart Autonomous Mobile Air Purifier for domestic and industrial air purification using AI and autonomous navigation. Raised **$32,742** for 2% equity from NIDHI PRAYAS (Government of India) and Forth Tech Limited; later exited, transferring the majority stake to Forth Tech.

Team Manager · Inspired Karters Gravity, BITS Pilani — Aug 2024 – Aug 2025

Managed the karting and Formula Student team of BITS Pilani (40+ students); designed and developed the FS vehicle with regenerative braking and secured **$10,000** in funding.

TEACHING EXPERIENCE

Computer Aided Design (ME F381) — Aug 2026 – present
Undergraduate Teaching Assistant · Department of Mechanical Engineering, BITS Pilani

Quality Control, Assurance and Reliability (ME F434) — Jan 2026 – May 2026
Undergraduate Teaching Assistant · Department of Mechanical Engineering, BITS Pilani

Mechanics of Solids (ME F211) — Aug 2025 – Dec 2025
Undergraduate Teaching Assistant · Department of Mechanical Engineering, BITS Pilani

Writing Coach · RISE Global — Aug 2025 – Mar 2026
Mentored 12+ high school students to write, execute and publish research papers in reputed journals.

HONOURS & AWARDS

NTSE Scholar (AIR 4), National Talent Search Examination, among 350,000+ candidates; centum in Mental Ability — 2021

KVPY Scholar (AIR 189), Kishore Vaigyanik Protsahan Yojana, among 200,000+ candidates; centum in Mathematics and Physics — 2022

Mathematics Olympiad (IOQM), AIR top 100 for three consecutive years — 2021 – 2023

European Rover Challenge 2025, CRISS Robotics · 3rd place globally — 2025

Australian Rover Challenge 2026, CRISS Robotics · first Indian team to qualify — 2026

International Rover Design Challenge 2025, conducted by SPROS, CRISS Robotics · 3rd place — Feb 2025

Project Prototype Presentation, APOGEE BITS Pilani, CRISS Robotics · 1st place — Apr 2024 & Mar 2025

3rd International Workshop on GESD, BITS Pilani · 2nd place in poster presentation — Sep 2024

ISIE Indian Karting Race 2024 — Oct 2024

CAD-a-Thon, APOGEE BITS Pilani · 2nd place — Apr 2023